division) and which did not require any advanced math concepts. As is the case with GSM8k, all problem solutions are positive integers[2]. No language models were used to construct this dataset.

To prevent data contamination concerns with GSM1k, we do not intend to release the dataset publicly at this time. However, we commit to releasing the full GSM1k dataset when at least one of the two following conditions have passed, whichever comes earlier. 1) Three open-source models with different pre-trained foundational model lineages reach 95% accuracy on GSM1k. 2) June 2025. At such a point, we believe that grade school mathematics will likely no longer be difficult enough to materially benchmark model releases and commit to releasing all data into the public domain under the MIT license. Additionally, to evaluate proprietary models, we were required to send over the dataset via API. Our belief is that model providers typically do not use such datapoints for model training. Nevertheless, in case GSM1k data is leaked through such means, we also hold out a small number of data points that have passed all quality checks but do not appear in the final GSM1k dataset. This data will also be released alongside GSM1k upon final release. We encourage future benchmarks to follow a similar pattern, where they are not released publicly lest they be gamed, but are precommitted to be released at a future date or upon a future condition. As part of this release, we will also open source our evaluation framework, which is based off of a fork of the LM Evaluation Harness by EleutherAI (Gao et al. [2023a]).

Finally, while we undertook extensive efforts to ensure maximum similarity between GSM8k and GSM1k, these results are only an approximation of an ideal world in which the test set of GSM8k was instead not publicly released and used for evaluations. We would recommend reading all results with the understanding that GSM8k and GSM1k are only highly similar, but not identically distributed despite all our efforts below.

3.1 Quality Checks

All questions passed through a total of 3 review layers. After initial creation, each task was manually reviewed by a subset of trusted annotators selected for strong past performance. These reviewers checked both for correctness as well as ensuring problems contained only grade school level math and proper formatting. To ensure that questions were answered correctly, we also do a second review layer by having an independent set of data annotators solve each question *without seeing the intended solution*. If this second solve produced a different answer to that of the initial solve, we discarded the problem. Finally, all problems were reviewed by a special team within Scale responsible for conducting general quality audits for data production. Out of a total of 2108 initial problems, 1419 passed the second solve stage and 1375 passed the general quality audit.

3.2 Matching the Difficulty Distribution of GSM8k

One important axis of recreating a benchmark is ensuring that new problems have a comparable difficulty to the original benchmark. To construct problems of difficulty N, we requested annotators to construct problems with N required resolution steps and prompted them with 3 examples from GSM8k with estimated difficulty N. The distribution of problems requested from annotators matched the estimated distribution in GSM8k. Difficulty is tricky to measure precisely, so we used an estimate based on the number of operations needed to solve the problem. This was extracted programmatically by counting the number of "calculator" tags in the problem solution. However, as not all problem solutions were formatted consistently, this estimate is only a rough estimate of actual difficulty. Additionally, the number of resolution steps in a problem does not necessarily directly correlate with the true level of problem difficulty.

Past work has also found that LLMs struggle with problems with larger numbers (Gao et al. [2023b]) even if they can solve otherwise identical problems with smaller numbers. To remove this as a potential confounding variable, our final processing step is to discard candidate problems from GSM1k so that the answer magnitude distributions of GSM8k and GSM1k are as similar as possible. This selection process is described in Figure 3. GSM1k consists of the 1205 problems that survive this final winnowing. Additionally, we run several checks to ensure that our efforts to match benchmark difficulty were successful.

[2]GSM8k has a few problems, likely errors, for which this is not the case.

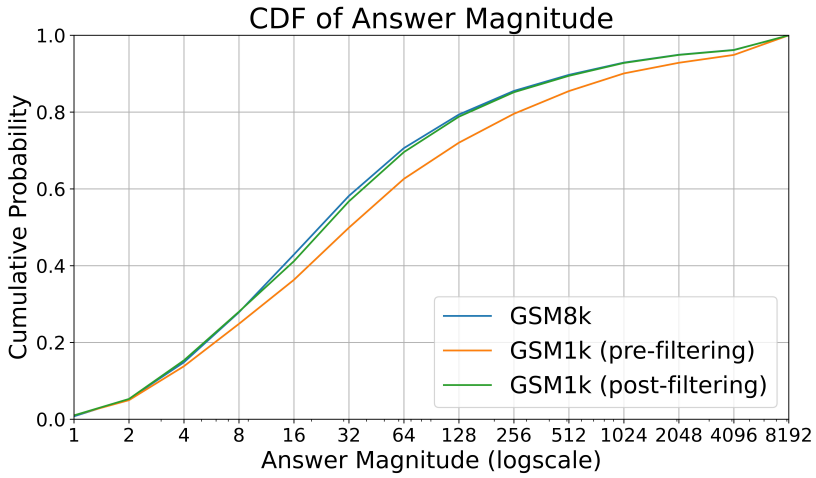

Figure 3: As the final step, we select 1205 problems to match the answer magnitude distribution of GSM8k as much as possible. The remaining problems are discarded and not included in the final dataset. Before discarding, we find that our generated problems tend to have slightly larger answers.

3.2.1 Human Differentiation Rates

The first test we run is human distinguishability. We present human annotators with a set of five questions, four of which were randomly selected from the original GSM8k dataset and one of which was selected from the newly created GSM1k dataset, and rewarded annotators for finding the odd one out. In an audit conducted using 19 annotators who were not involved in the problem creation process, we found that annotators were able to correctly identify the lone GSM1k example 21.83% of the time out of 1205 attempts (20% is pure chance). Separately, we also tested several paper authors who had not yet seen the data and they were also unable to perform much better than random. This suggests minimal differences between GSM8k and GSM1k, at least as measured by the human eye.

3.2.2 Human Solve Rates

To ensure similar solve rates, we also asked annotators to solve questions under time pressure. 14 annotators who had not participated in the problem creation process attempted to solve as many GSM8k problems as they could in 15 minutes and were rewarded based on the number of problems they solved. We repeated this exact setup for GSM1k. Annotators were able to solve an average of 4.07 ± 0.93 problems on the GSM8k dataset. They were able to solve 4.36 ± 1.11 problems on the GSM1k dataset, where the error rates are the standard deviations of the evaluation. This suggests that GSM1k is comparable in difficulty (and perhaps even slightly easier) than GSM8k. As such, substantial decreases in model accuracy on GSM1k compared to GSM8k are likely not explainable due to differences in dataset difficulty.

3.2.3 LLM Solve Rates

Finally, we sanity check our results by measuring solve rates of several models that are known to not be contaminated by GSM8k due to being released before the publication of the GSM8k dataset. Due to the relative scarcity of LLMs trained only on pre-2021 data, we evaluate only GPT-NeoX-20B (Black et al. [2022]) and GPT-2 (Radford et al. [2019]). For these two language models, we find minimal difference between their solve rates of GSM8k and GSM1k (Figure 12).

4 Results

To evaluate models, we use a fork of EleutherAI's LM Evaluation Harness with minor modifications. We use the default settings for evaluation, except for increasing the maximum number of allowed generated tokens from 256 to 1000, as we notice that the default setting did not allow some models to

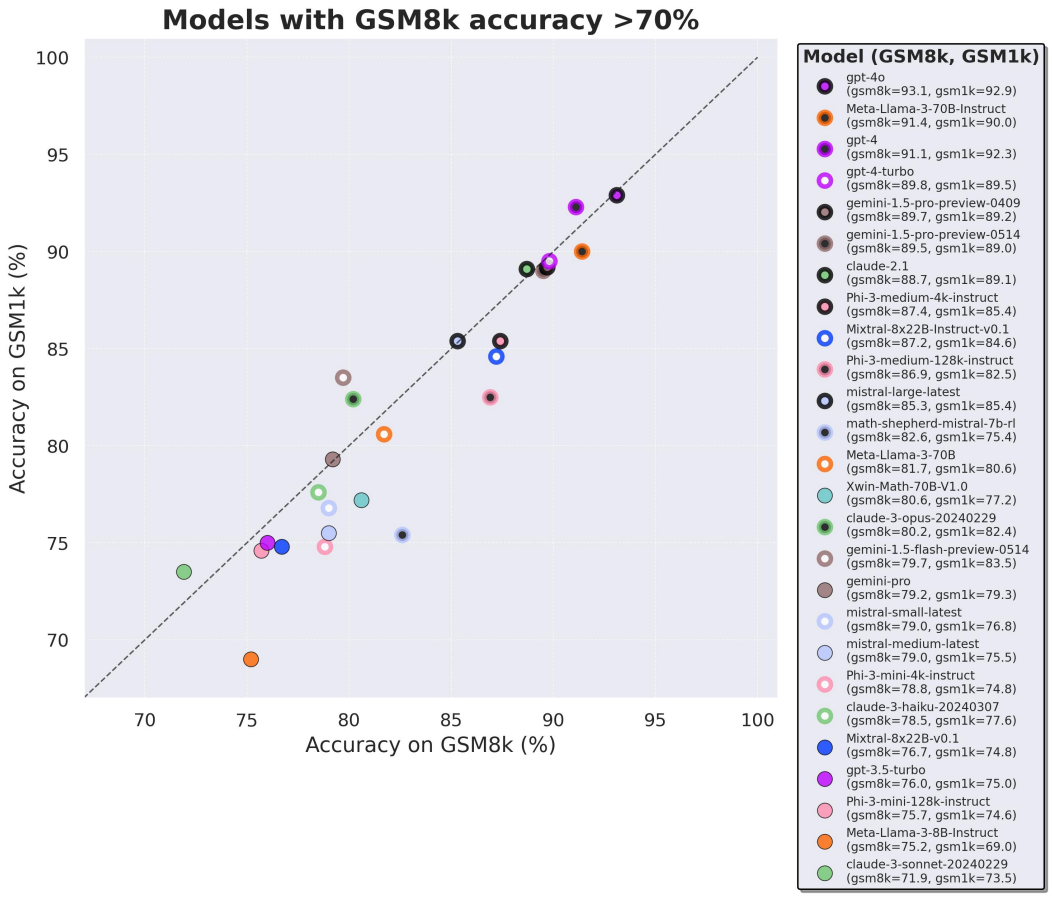

Figure 4: Models with over 70% accuracy on GSM8k compared to the line of no overfit. This plot is zoomed into the relevant sections (70-100% accuracy). Note that some models, especially the Claude family, perform above the 45 degree line, which is consistent with our findings in Section 3 that GSM1k is slightly easier than GSM8k. In contrast, many other models lie well below this line.

complete their full chain-of-thought reasoning before being truncated. Both GSM8k and GSM1k questions are run with the same prompt of using 5 randomly drawn examples from the GSM8k train set, as is standard in the field. An example prompt is provided in Appendix D. All open-source models are evaluated at temperature 0 for reproducibility. For open source models, we use vLLM to speed up model inference if a model is compatible with the library. Otherwise, we default to inference using standard HuggingFace libraries. Closed-source models were queried through the LiteLLM library which unifies the API call format for all proprietary models evaluated. All API model results were from queries between April 16 - July 10, 2024 and use the default settings.

LM Evaluation Harness uses an automatic evaluation method which extracts the last numeric answer in the response and compares this to the correct answer. However, in some cases, models will produce "correct" answers in a format that do not match the given examples, resulting in their answers being marked as incorrect. To explore the effect of this on the results, we run an ablation where we select a subset of models and use human annotation to manually extract answers that are not correctly formatted (Appendix J). We do not find major changes in our findings for the models examined.

As model benchmark performance is highly dependent on choice of prompt and evaluation setting, our reported GSM8k numbers may occasionally be below the reported model benchmark numbers, as we use a standardized setting for all models instead of the prompt that maximizes each individual model's performance. Additionally, we explore the effect of different prompt formulations with several ablations. In Appendix E, we report results with an alternative prompting format that uses non-GSM8k examples as n-shot examples and a slightly different answer phrasing. Additionally, we explore the effect of varying the number and source of the n-shot examples used in Appendix K and